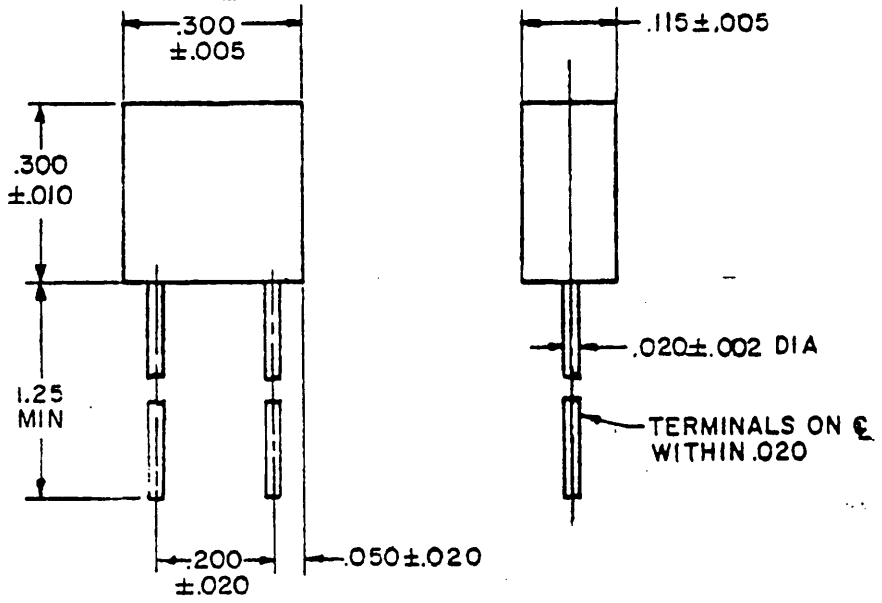
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, GLASS DIELECTRIC.

STYLES CY07 AND CY07.....TX

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-11272.

INACTIVE FOR DESIGN AFTER DATE OF THIS SPECIFICATION. USE MIL-C-23269.



INCHES	MM	INCHES	MM	INCHES	MM
.002	.05	.020	.51	.200	5.08
.005	.13	.050	1.27	.300	7.62
.010	.25	.115	2.92	1.25	31.75

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest . 01 mm) are given for general information only and are based upon 1 inch = 25. 4 mm.

FIGURE 1. Capacitors, Fixed, Glass Dielectric, Styles CY07 and CY07. . . . TX.

FSC 5910

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REQUIREMENTS:

Design and construction:

Dimensions and configuration – See figure.

Material:

Dielectric – Glass.

Case – Epoxy.

Lead – Gold plated dumet (MIL-STD-1276, type D).

Capacitance value – See table I.

TABLE I. *Styles CY07 and CY07.....TX.*

Type designation1/	Capacitance (pF)	DC rated voltage	Capacitance tolerances available (±)
CY07C621-	620	300	FGJ
CY07C621-TX	620	300	FGJ
CY07C681-	680	300	FGJ
CY07C681-TX	680	300	FGJ
CY07C751-	750	300	FGJ
CY07C751-TX	750	300	FGJ
CY07C821-	820	300	FGJ
CY07C821-TX	820	300	FGJ
CY07C911-	910	300	FGJ
CY07C911-TX	910	300	FGJ
CY07C102-	1000	300	FGJ
CY07C102-TX	1000	300	FGJ

1/Complete type designation will include an additional letter symbol to indicate capacitance tolerance, where applicable, and the letters TX for TX version parts.

Examples: Non-TX version – CY07C102J
TX version – CY07C102JTX.

Capacitance tolerance – See table I.

DC rated voltage – 300 volts.

Temperature range – -55° to +125°C.

Dielectric withstanding voltage: Method 301 of MIL-STD-202 – 200 percent dc rated voltage for not less than 1 second nor more than 5 seconds.

Barometric pressure: Method 105 of MIL-STD-202.

Test condition B (50,000 ft).

150 percent dc rated voltage for not less than 1 second nor more than 5 seconds.

Insulation resistance: Method 302 of MIL-STD-202.

At 25°C – 100,000 megohms, minimum.

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Capacitance: Method 305 of MIL-STD-202.

Dissipation factor: Not to exceed 0.001.

Quality factor (Q):
Method 306 of MIL-STD-202.
Test frequency – 1 MHz ±200 kHz.
Not less than 1,000.

Shock (specified pulse): Method 213 of MIL-STD-202, test condition I (100 G).

Vibration, high frequency: Method 204 of MIL-STD-202, test condition B (15G).

Temperature cycling: Method 102 of MIL-STD-202, test condition C.

Immersion: Method 104 of MIL-STD-202, test condition B.
Dielectric withstanding voltage – No breakdown.
Insulation resistance – Not less than initial requirement.
Capacitance – Change not more than 0.5 percent of nominal value from initial value.
Dissipation factor – Not to exceed 0.001.

Moisture resistance: Method 106 of MIL-STD-202:
At 90 to 95 percent relative humidity.
Insulation resistance – Not less than 1,000 megohms.
At 50 ±5 percent relative humidity.
Dielectric withstanding voltage – No breakdown.
Insulation resistance – Not less than initial requirement.
Capacitance – Change not more than 0.5 percent of nominal value from initial value.
Dissipation factor – Not to exceed 0.001.

Life: 150 percent dc rated voltage at 125° +4,-0°C for 2,000 hours.
Group A test – 250 hours; group B continuation test for 1,750 hours.
For qualification test:
At 25°C:
Insulation resistance – Not less than initial requirement.
Capacitance – Change not more than 0.5 percent of nominal value from initial value.
Dissipation factor – Not to exceed 0.0025.
At 125°C:
Insulation resistance – Not less than 10,000 megohms.
Capacitance – Change not more than 0.5 percent of nominal value from initial 125°C value.
Dissipation factor – Not to exceed 0.007.

For performance check:
Insulation resistance – Not less than initial requirement.
Capacitance – Change not more than 0.5 percent of nominal value from initial value.
Dissipation factor – Not to exceed 0.0025.

For continuation test: Capacitors shall meet requirements specified for qualification test.

Temperature coefficient: 140 ±25 parts/million/°C.

Capacitance drift: 0.1 percent.

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QUALITY ASSURANCE PROVISIONS

Sampling and inspection: MIL-C-11272 and as specified herein.

Qualification inspection: MIL-C-11272 on the respective non-TX version. (The TX version shall be procured from sources listed in QPL-11272 for the respective non-TX version.)

Quality conformance inspection: MIL-C-11272 and as follows:

Group A inspection: For the non-TX version, MIL-C-11272. For the TX version, 100 percent of the capacitors in a lot shall be subjected to the initial tests, power-stress conditioning, and post-stress conditioning measurements listed below. Visual and mechanical examinations shall be performed to 1.0 percent major and 2.5 percent minor AQLs. Successful performance of the barometric pressure and life tests on the respective non-TX version will suffice to demonstrate conformance to the requirement for TX version.

Group B inspection: MIL-C-11272: Successful performance of group B inspection for the respective non-TX version will suffice to demonstrate conformance to the requirement for TX version.

Initial tests: IR, capacitance, and DF shall be measured. Items that are out of capacitance tolerance, DF, and IR limits shall be removed from the lot.

Power-stress conditioning: Each capacitor intended for delivery as a TX part shall be subjected continuously to five times the rated voltage for 50 hours at an ambient temperature of 25°C.

Post power-stress conditioning measurement: After conditioning, each capacitor shall be measured for IR, capacitance, and DF. Items that fail to meet the initial requirements shall be removed from the inspection lot, and the quantity removed noted on the permanent lot history. Only lots having not more than 5 percent rejects or one capacitor, whichever is greater, as a result of this test shall be approved for delivery.

Acceptance: Acceptance of TX parts shall be based on the qualification, quality conformance inspection, sampling criteria of MIL-C-11272, and power-stress conditioning.

Certification: Each lot delivered as TX parts shall be accompanied by a certificate verifying that power-stress conditioning has been performed and satisfactorily met.

Traceability: Each lot intended for delivery as TX parts shall be assigned a unique lot number prior to any tests. The lot number shall identify its date of manufacture. The manufacturer shall maintain a chronological file of these lot numbers and the data records shall be made available on request of the procuring activity. The data records shall be retained by the supplier for a period of 3 years.

Qualification: MIL-C-11272. Information pertaining to qualification of products may be obtained from the Defense Electronics Supply Center, (DESC-E), 1507 Wilmington Pike, Dayton, Ohio 45401.

Marking: Non-TX parts in accordance with MIL-C-11272. TX parts shall be marked as follows:

Example:

CY07C
102J
TX

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Custodian:
Air Force - 11

Review activities:
Air Force - 17, 19, 85

Preparing activity:
Air Force - 11

Agent:
DSA - ES

(Project 5910-F160-1)

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b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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